

Soyab Karki

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EDUCATION

Mississippi State University

Bachelor of Science in Computer Science, Minor in Data Science and Mathematics

Starkville, MS

Aug. 2023 – May 2027

- **GPA:** 4.0/4.0
- **Shackoul's Honors College**
- **Relevant Coursework:** Operating Systems, Data Structures & Algorithms, Computer Organization, Parallel Algorithms, Software Development, Data Science, Data Visualization, Machine Learning, AI Robotics

EXPERIENCE

Data Analytics Research Assistant

Mississippi State University

Aug. 2024 – Present

Starkville, MS

- Developed data pipelines for 300GB+ consumer review datasets using batch processing and Parquet storage, improving data processing speed and storage consumption by 10x.
- Automated filtering of 2,500+ research papers using OpenAI API and custom keyword matching in Python, reducing manual review time from days to minutes.
- Built RAG-based AI agents using LangChain to automate metadata analysis for research papers.
- Co-authored "Modeling consumer ratings: The trade-off between quality and consumer preference," published in the European Journal of Operational Research.

Systems Software Engineer Fellow

High Performance Computing Collaboratory

Jan. 2024 – Present

Starkville, MS

- Parallelized 10+ algorithmic problems using MPI in C++, achieving up to 5x speedup over sequential code.
- Integrated a custom GoLang and SQLite middleware layer for an in-house LLM chat service to manage real-time token tracking and user authentication.
- Deployed Kubernetes using K3s on a lightweight 3-node Linux cluster to prototype HPC workflows.

Data Scientist Intern

United States Department of Agriculture - Agricultural Research Service

May 2024 – July 2024

Starkville, MS

- Developed 6 risk-analysis ML models using Scikit-Learn in Python, achieving baseline accuracy of up to 97%.
- Processed 1.5 years of meteorological data from Google Earth Engine using Python API for ML training.
- Presented findings at 3 regional conferences, clearly communicating complex AI workflows to diverse audiences.

TECHNICAL SKILLS

Languages: Python, Go, C/C++, JavaScript/Typescript, SQL

Backend & Frontend: FastAPI, React.js, Express.js, Node.js, PostgreSQL, Redis, HTML/CSS

Data & MLOps: Airflow, Spark, Kafka, PyTorch, Pandas, Scikit-Learn, LangChain

Cloud & DevOps: Docker, Kubernetes, AWS, Git, GitHub, Prometheus, Grafana

PROJECTS

Distributed LLM-As-A-Service | *Kubernetes, FastAPI, React.js, NGINX, Github Actions*

Jan. 2025

- Architected a horizontally-scalable microservice platform on Kubernetes, implementing an API gateway to route and load balance requests to distributed LLM workers.
- Developed a full-stack AI chat application using React.js and an event-driven backend, automating deployments using CI/CD pipeline for zero-downtime updates.

Reddit Data Pipeline | *Python, Airflow, PostgreSQL, Docker, Prometheus, Grafana*

Aug. 2025

- Built a containerized ETL pipeline that pulls stock discussions from 3 Reddit forums, runs FinBERT sentiment analysis, enriches with news/stock data, and stores in PostgreSQL with materialized views for fast queries.
- Designed real-time dashboards using Prometheus and Grafana to improve system observability.

Resume Radar | *FastAPI, Redis, React.js, Chrome Extension API, Docker*

July 2025

- Built a Chrome extension that analyzes resumes against job descriptions using OpenAI API.
- Optimized FastAPI backend with REST APIs and Redis caching, reducing system latency by over 90%.

LEADERSHIP

Data Science Club at MSU | *President*

Feb. 2024 – Present

- Recruited 50+ members and raised \$3,000+ to run datathons and workshops.

ACM SIG-HPC | *Founding Member*

Sep. 2023 – Present

- Configured mini HPC clusters using Raspberry Pi; competed in 2 major supercomputing competitions.